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ELECTRONIC DEVICE INCLUDING A TOUCH DETECTION SYSTEM AND A NOISE DETECTION SYSTEM

Abstract

An electronic device includes a touch detection system and a noise detection system. The touch detection system includes a first capacitive sensing channel, and a touch detection circuitry to perform a series of touch detections based on a capacitance on the first capacitive sensing channel. The noise detection system includes a second capacitive sensing channel, and a noise detection circuitry to perform at least one noise detection measurement, wherein a respective noise detection measurement comprises determining a respective noise-related capacitance measure associated with a capacitance on the second capacitive channel. The noise detection system to determine a presence of a noise condition based on the at least one noise detection measurement, and in response to determining the noise condition, generate a touch detection inhibit signal to inhibit at least one touch detection in the series of touch detections.

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Background/Summary

RELATED APPLICATION

[0001] This application claims priority to commonly owned Chinese Patent Application No. 202410250784.7 filed Mar. 5, 2024, the entire contents of which application are hereby incorporated by reference for all purposes.

BACKGROUND

[0002] A conventional mutual capacitance sensing system touch detection includes one or more sensing channels for detecting a variance in capacitance, which may indicate a touch, for example a touch on a touchscreen or other touch device. To measure the capacitance on each channel, a burst wave may be transmitted from a transmit pin (referred to as an “X pin”) and a resulting signal may be measured (sampled) at a corresponding receiving pin (referred to as a “Y pin”). A respective X-Y pin pair defines a respective sensing channel of the relevant touch device (e.g., touchscreen).

[0003] Conventional capacitance sensing systems, e.g., touch detection systems described above, are commonly affected by noise, which may provide erroneous touch detection results. Noise may include, for example, voltage ripple, radiation emissions, and electrostatic discharge proximate the capacitance sensing system, without limitation.

[0004] There is a need for an improved capacitance sensing system, e.g., for use in a touchscreen or other touch device.

SUMMARY

[0005] The present disclosure provides an electronic device including a touch detection system, e.g., to detect touches on a touchscreen or other touch device, and a noise detection system to detect noise conditions, and in response, to generate touch detection inhibit signals to inhibit touch detection(s) of the touch detection system, for example to prevent erroneous (e.g., false positive) touch detections.

[0006] One aspect provides an electronic device including a touch detection system and a noise detection system. The touch detection system includes a first capacitive sensing channel, and a touch detection circuitry to perform a series of touch detections based on a capacitance on the first capacitive sensing channel. The noise detection system includes a second capacitive sensing channel, and a noise detection circuitry to perform at least one noise detection measurement, wherein a respective noise detection measurement comprises determining a respective noise-related capacitance measure associated with a capacitance on the second capacitive channel, determine a presence of a noise condition based on the at least one noise detection measurement, and in response to determining the noise condition, generate a touch detection inhibit signal to inhibit at least one touch detection in the series of touch detections.

[0007] In some examples, the electronic device includes a sense electrode, a touch detection electrode, and a noise detection electrode, wherein the first capacitive sensing channel includes a capacitive coupling between the touch detection electrode and the sense electrode, and wherein the second capacitive sensing channel includes a capacitive coupling between the noise detection electrode and the sense electrode.

[0008] In some examples, performing a respective touch detection in the series of touch detections comprises detecting a touch-related capacitance measure at the sense electrode at a first time, and performing the respective noise detection measurement comprises detecting the respective noise-related capacitance measure at the sense electrode at a second time distinct from the first time.

[0009] In some examples, the electronic device comprises an IC package including an IC die including the touch detection circuitry, the noise detection circuitry, and a plurality of IC die pads including a first IC die pad, a second IC die pad, and a third IC die pad. The IC package includes a plurality of IC package pins, including a first IC package pin connected to the first IC die pad, and a second IC package pin connected to the second IC die pad. The sense electrode includes the first IC die pad and the first IC package pin connected to the first IC die pad, the wherein the touch detection electrode includes the second IC die pad and the second IC package pin connected to the second IC die pad, and the noise detection electrode comprises the third IC die pad.
[0010] In some examples, the noise detection electrode comprises the third IC die pad and a conductive trace extending from the third IC die pad.
[0011] In some examples, the plurality of IC package pins includes a third IC package pin connected to the third IC die pad, and the noise detection electrode comprises the third IC die pad and the third IC package pin connected to the third IC die pad.
[0012] In some examples, the sense electrode includes a first touch electrode element connected to the first IC die pad, the touch detection electrode includes a second touch electrode element connected to the second IC die pad, and the noise detection electrode includes a third touch electrode element connected to the third IC die pad.
[0013] In some examples, the electronic device includes a touch detection electrode, a first sense electrode, a noise detection electrode, and a second sense electrode, wherein the first capacitive sensing channel includes a capacitive coupling between the touch detection electrode and the first sense electrode, and wherein the second capacitive sensing channel includes a capacitive coupling between the noise detection electrode and the second sense electrode.
[0014] In some examples, determining the presence of the noise condition based on the at least one noise-related capacitance measurement comprises comparing the at least one noise-related capacitance measure to a noise detection threshold.
[0015] In some examples, generating the touch detection inhibit signal to inhibit the at least one touch detection comprises generating the touch detection inhibit signal to prevent a performance of the at least one touch detection.
[0016] In some examples, generating the touch detection inhibit signal to inhibit the at least one touch detection comprises generating the touch detection inhibit signal to inhibit a usage of the at least one touch detection.
[0017] In some examples, generating the touch detection inhibit signal to inhibit the at least one touch detection comprises generating the touch detection inhibit signal to inhibit a predefined multiple number of touch detections.
[0018] In some examples, the noise detection circuitry to (a) perform at least one further noise detection measurement, wherein a respective further noise detection measurement comprises determining a respective further noise-related capacitance measure associated with the capacitance on the second capacitive channel, (b) determine an absence of the noise condition based on the at least one further noise detection measurement, and (c) in response to the determination of the absence of the noise condition, not generate the touch detection inhibit signal.
[0019] In some examples, the touch detection circuitry to perform a respective touch detection in the series of touch detections by determining a capacitance measure associated with a capacitance on the first capacitive sensing channel, and comparing the touch-related capacitance measurement to a touch detection threshold.
[0020] In some examples, the device comprises a touch device including a first touch electrode element and a second touch electrode element, and the first capacitive sensing channel includes a capacitive coupling between the first touch electrode element and the second touch electrode element.
[0021] In some examples, the electronic device comprises an IC die including the touch detection circuitry and the noise detection circuitry, and a touch device including a first touch electrode

element connected to a first IC pad of the IC die, and a second touch electrode element connected to a second IC pad of the IC die and capacitively coupled to the first touch electrode element. The first touch electrode element and the second touch electrode element are arranged along the first capacitive sensing channel, and the touch detection circuitry to perform a series of touch detections based on a capacitive coupling between the first touch electrode element and the second touch electrode element.

[0022] One aspect provides a method including performing, using a touch detection circuitry provided in an integrated circuit, at least one first touch detection based on a capacitance on a first capacitive sensing channel; performing, using a noise detection circuitry provided in the IC device, at least one noise detection measurement based on a capacitance on a second capacitive channel; determining, by the noise detection circuitry, a presence of a noise condition based on the at least one noise detection measurement; and in response to determining the presence of the noise condition, inhibiting at least one second touch detection.

[0023] In some examples, performing the at least one noise detection comprises determining at least one noise-related capacitance measure associated with the capacitance on the second capacitive channel, and determining the presence of the noise condition comprises comparing the at least one noise-related capacitance measure to a noise detection threshold.

[0024] In some examples, inhibiting at least one second touch detection comprises preventing a performance of at least one second touch detection by the touch detection circuitry.

[0025] In some examples, inhibiting at least one second touch detection comprises preventing a predefined number of second touch detections or preventing second touch detections for a predefined period of time.

[0026] In some examples, inhibiting at least one second touch detection comprises preventing second touch detections until a determination, by the noise detection circuitry, that the noise condition is absent.

[0027] In some examples, the determination, by the noise detection circuitry, that the noise condition is absent comprises a determination, by the noise detection circuitry, that the noise condition is not present for a predefined minimum number of noise detection measurements.

[0028] In some examples, inhibiting at least one second touch detection comprises preventing or inhibiting a use of at least one second touch detection performed by the touch detection circuitry.

[0029] In some examples, performing at least one first touch detection comprises detecting at least one first touch at a touch device, and inhibiting at least one second touch detection comprises (a) inhibiting a detection of at least one second touch at the touch device or (b) inhibiting a use of at least one detected second touch at the touch device.

Description

DESCRIPTION OF THE DRAWINGS

[0030] Example aspects of the present disclosure are described below in conjunction with the figures, in which:

[0031] FIG. **1** shows an example electronic device including a touch detection system for capacitive touch detection, and a noise detection system for detecting noise conditions and controlling the operation of the electronic device in response to detected noise conditions;

[0032] FIG. **2** shows an example implementation of the electronic device shown in FIG. **1**, including a touch detection channel and a noise detection channel sharing a common electrode;

[0033] FIG. **3** shows another example implementation of the electronic device shown in FIG. **1** including a touch detection channel and a separate noise detection channel;

[0034] FIG. **4** shows an example implementation of the example electronic device shown in FIG. **2**;

[0035] FIG. **5** shows another example implementation of the example electronic device shown in

FIG. **2**;

[0036] FIG. **6** shows another example implementation of the example electronic device shown in FIG. **2**;

[0037] FIG. **7** shows an example electronic device including multiple touch detection channels;

[0038] FIG. **8** shows an example method for managing capacitive touch detection, e.g., implemented by any of the example electronic devices disclosed herein; and

[0039] FIG. **9** shows another example method for managing capacitive touch detection, e.g., implemented by any of the example electronic devices disclosed herein.

[0040] It should be understood that the reference number for any illustrated element that appears in multiple different figures has the same meaning across the multiple figures, and the mention or discussion herein of any illustrated element in the context of any particular figure also applies to each other figure, if any, in which that same illustrated element is shown.

DETAILED DESCRIPTION

[0041] FIG. **1** shows an example electronic device **100** including a touch detection system **102** (e.g., a mutual capacitance touch detection system) for capacitive touch detection and a noise detection system **104** for detecting noise conditions and controlling the operation of the electronic device **100** in response to detected noise conditions.

[0042] The touch detection system **102** includes a first capacitive sensing channel **103** and a touch detection circuitry **112** to perform a series of touch detections **120** based on a capacitance on the first capacitive sensing channel **103**. The first capacitive sensing channel **103** may also be referred to as a “touch detection channel.” In some examples, a respective touch detection **120** may include determining a capacitance measure (e.g., at least one voltage or current, without limitation) associated with a capacitance on the first capacitive sensing channel **103**, comparing the capacitance measure to a touch detection threshold, and detecting a touch if the capacitance measure exceeds the touch detection threshold.

[0043] The noise detection system **104** includes a second capacitive sensing channel **105**, and a noise detection circuitry **114** to (a) perform noise detection measurement(s) **130** to determine noise-related capacitance measure(s) **132** associated with a capacitance on the second capacitive sensing channel **105**, (c) determine a presence (or absence) of a noise condition **134** based on the noise-related capacitance measure(s) **132**, and in response to determining the noise condition **134** is present, generate a touch detection inhibit signal **136** to inhibit at least one touch detection **120**, for example to prevent or inhibit the performance or usage of at least one touch detection **120** during noisy periods (i.e., periods in which the noise condition **134** is present). The second capacitive sensing channel **105** may also be referred to as a “noise detection channel.”

[0044] As used herein, a “capacitive sensing channel” (e.g., first capacitive sensing channel **103** or second capacitive sensing channel **105**, without limitation) refers to a circuit including at least two capacitively coupled electrodes, wherein the circuit is connected to respective circuitry (e.g., touch detection circuitry **112** or noise detection circuitry **114**, without limitation) for detecting, measuring, or otherwise sensing a capacitance between the electrodes, which capacitance is also referred to herein as the capacitance on the respective capacitive sensing channel.

[0045] As discussed in more detail below, in some examples the first capacitive sensing channel **103** may comprise a capacitive coupling between a touch detection electrode and a sense electrode, and the second capacitive sensing channel **105** comprise a capacitive coupling between a noise detection electrode and the same sense electrode or another sense electrode. In the various example disclosed herein, a device may perform either active or passive touch detection on the respective first capacitive sensing channel **103** (touch detection channel), and may perform either active or passive noise detection on the respective second capacitive sensing channel **105** (noise detection channel).

[0046] In examples that utilize active touch detection on the first capacitive sensing channel **103** (touch detection channel), the touch detection electrode may comprise a “transmit” or “drive” electrode over which electrical signals (e.g., wave bursts) are transmitted, and the sense electrode may comprise a "receive" or other electrode at which resulting electrical signals (e.g., voltage or current) are sensed. In examples that utilize passive touch detection on the first capacitive sensing channel **103**, the touch detection electrode and sense electrode respectively may comprise passive electrodes, wherein electrical signals (e.g., voltage or current) may be sensed at one or both passive electrodes and analyzed for touch detection.

[0047] Similarly, in examples that utilize active noise detection on the second capacitive sensing channel **105** (noise detection channel), the noise detection electrode may comprise a "transmit" or "drive" electrode over which electrical signals (e.g., wave bursts) are transmitted, and the sense electrode may comprise a "receive" or other electrode at which resulting electrical signals (e.g., voltage or current) are sensed. In examples that utilize passive noise detection on the second capacitive sensing channel **105**, the noise detection electrode and sense electrode respectively may comprise passive electrodes, wherein electrical signals (e.g., voltage or current) may be sensed at one or both passive electrodes and analyzed for noise detection.

[0048] As discussed below, the touch detection electrode, noise detection electrode, and sense electrode(s) may respectively include (a) conductive elements of an integrated circuit (IC) die (e.g., IC die pads and/or conductive traces), (b) conductive elements of an IC package in which the IC die is mounted (e.g., IC package pins wire bonded or otherwise connected to respective IC die pads), and/or (c) electrode elements provided in a touch device, e.g., a touchscreen (e.g., provided in a smartphone, laptop, tablet, or other device), a touch button or switch, a touch pad, a touch slider, a touch surface, or other touch device. The term pin, as used herein, is not meant to be limited to any particular type of physical structure, and may include, without limitation, gull-wing or J-lead terminals, solder balls, or lands.

[0049] As discussed above, noise detection circuitry **114** may include circuitry to perform noise detection measurement(s) **130** to determine noise-related capacitance measure(s) **132** associated with a capacitance on the second capacitive sensing channel **105**, and determine a presence (or absence) of a noise condition **134** based on the noise-related capacitance measure(s) **132**. Noise-related capacitance measure(s) **132** determined by the noise detection circuitry **114** may include voltage, current, or other electrical property influenced by noise in the vicinity of the electronic device **100**, for example, voltage ripple, radiation emissions, or electrostatic discharge near the electronic device **100**.

[0050] In examples in which the first capacitive sensing channel **103** and second capacitive sensing channel **105** share a common IC die pad, both the touch detection circuitry **112** and noise detection circuitry **114** may be connected to the shared IC die pad for (a) performing respective touch detections **120** (by the touch detection circuitry **112**) and (b) performing respective noise detection measurement(s) **130** (by the noise detection circuitry **114**). Touch detections **120** and noise detection measurement(s) **130** using the shared IC die pad may be performed at different times, e.g., in an alternating manner. In examples in which the second capacitive sensing channel **105** is separate from the first capacitive sensing channel **103**, the noise detection circuitry **114** may perform respective noise detection measurement(s) **130** (e.g., at a designated IC die pad) at any time relative to respective touch detections **120** performed by the touch detection circuitry **112**.

[0051] The noise detection circuitry **114** may determine the presence (or absence) of the noise condition **134** by comparing one or more noise-related capacitance measure(s) **132** with one or more noise detection thresholds, e.g., a threshold magnitude of voltage, current, or other electrical property. For example, noise detection circuitry **114** may compare a noise-related capacitance measure **132** with a noise detection threshold, and either (a) determine the presence of the noise condition **134** if the noise-related capacitance measure **132** exceeds the threshold, or alternatively (b) determine the absence of the noise condition **134** if the noise-related capacitance measure **132** does not exceed the threshold. In some examples, a noise detection protocol implemented in the noise detection circuitry **114** may specify a determination that the noise condition **134** is present when a respective (e.g., single) noise-related capacitance measure **132** is determined to exceed the threshold. In other examples, a noise detection protocol implemented in the noise detection circuitry **114** may specify a determination that the noise condition **134** is present when a defined minimum consecutive number of noise-related capacitance measures **132** are determined to exceed the threshold (or alternatively, when a defined minimum number of noise-related capacitance measures **132** are determined to exceed the threshold within a defined time period), depending on the respective noise detection protocol implemented in the noise detection circuitry **114**. A determination by the noise detection circuitry **114** that the noise condition **134** is not present may be considered herein as a determination the noise condition **134** is absent.

[0052] Noise detection threshold(s) may be stored in memory of the noise detection circuitry **114** or in separate memory accessible by the noise detection circuitry **114**. In some examples, noise detection circuitry **114** may utilize noise detection thresholds, e.g., specifying different magnitudes of voltage, current, or other electrical property, wherein exceeding the different thresholds may trigger different responsive actions or generation of different control signals. In some examples, including examples that implement (a) active touch detection via first capacitive sensing channel **103** (e.g., wherein the touch detection circuitry **112** transmits signals (e.g., wave bursts) on a touch detection electrode) and (b) passive noise detection, a noise detection threshold used by the noise detection circuitry **114** may have a lower magnitude than a touch detection threshold used by the touch detection circuitry **112**. In some examples, the noise detection and touch detection threshold may have the same magnitude.

[0053] As discussed above, in response to determining the presence of the noise condition **134** (e.g., a determination of respective noise-related capacitance measure(s) exceeding a noise detection threshold), the noise detection circuitry **114** may generate a touch detection inhibit signal **136** to touch detection circuitry **112** so as to inhibit at least one touch detection **120**. Alternatively, if the noise detection circuitry **114** determines the noise condition **134** is absent, the noise detection circuitry **114** does not generate the touch detection inhibit signal **136**.

[0054] In some examples, the touch detection inhibit signal **136** to inhibit at least one touch detection **120** may comprise a signal to inhibit a single touch detection **120**, to inhibit a predefined number of touch detections **120**, or to inhibit touch detections **120** for a predefined period of time. In other examples, the touch detection inhibit signal **136** may inhibit touch detections **120** until the determination (by the noise detection circuitry **114**) of a defined event to continue (re-start) touch detections **120**. Such defined event may comprise, for example, a defined consecutive number of noise detection determinations indicating the noise condition **134** is absent (e.g., when a defined consecutive number of noise-related capacitance measures **132** are determined not to exceed the noise detection threshold). For example, after determination of the noise condition **134**, the touch detection inhibit signal **136** may initiate a debouncing process in which a noise-absence detection counter must reach a defined value (wherein the counter advance after each successive determination of an absence of the noise condition **134**) before continuing (re-starting) touch detections **120**.

[0055] In some examples, the touch detection inhibit signal **136** to inhibit at least one touch detection **120** comprises signaling communicated to the touch detection circuitry **112** to interrupt (i.e., prevent a performance of) at least one touch detection **120** that would otherwise be performed by the touch detection circuitry **112**. In other examples, the touch detection inhibit signal **136** to inhibit at least one touch detection **120** comprises signaling communicated to the touch detection circuitry **112** to prevent or inhibit a usage of at least one touch detection **120** performed by the touch detection circuitry **112**. For example, the touch detection inhibit signal **136** may allow the touch detection circuitry **112** to continue performing touch detections **120** without interruption, but may prevent or otherwise inhibit the use of the results of such touch detections **120**, e.g., by preventing the touch detection circuitry **112** from signaling a detected touch to a responsive electronic device.

[0056] FIG. **2** shows an example electronic device **200** representing an example implementation of the electronic device **100** shown in FIG. **1** and discussed above. The example electronic device **200** includes (a) the touch detection system **102** including the first capacitive sensing channel **103** and touch detection circuitry **112** to perform touch detections **120** based on the capacitance on the first capacitive sensing channel **103**, and (b) the noise detection system **104** including the second capacitive sensing channel **105** and noise detection circuitry **112** to detect noise conditions **134** and to control the operation of the electronic device **100** in response to detected noise conditions **134**, e.g., as discussed above.
[0057] As shown in FIG. **2**, the first capacitive sensing channel **103** connected to the touch detection circuitry **112** includes a touch detection electrode **108** capacitively coupled to a sense electrode **106**, and the second capacitive sensing channel **105** connected to the noise detection circuitry **114** includes a noise detection electrode **110** capacitively coupled to the sense electrode **106**. Thus, the sense electrode **106** is shared by the first capacitive sensing channel **103** and the second capacitive sensing channel **105**.
[0058] The sense electrode **106**, touch detection electrode **108**, and noise detection electrode **110** may respectively include (a) conductive elements of an IC die (e.g., IC die pads and/or conductive traces), (b) conductive elements of an IC package in which the IC die is mounted (e.g., IC package pins wire bonded or otherwise connected to respective IC die pads), and/or (c) electrode elements provided in a touch device. In some examples (e.g., as shown in FIGS. **4-7** discussed below), the sense electrode **106** comprises an IC die pad or IC package pin connected to both the touch detection circuitry **112** and the noise detection circuitry **114**. The touch detection circuitry **112** and the noise detection circuitry **114** may perform respective measurements (e.g., detecting voltage, current, or other electrical parameter) at the shared sense electrode **106** at different times (e.g., in an alternating manner), for example by selectively connecting the shared sense electrode **106** to the touch detection circuitry **112** and the noise detection circuitry **114** (e.g., using suitable switching circuitry).
[0059] In some examples, e.g., examples that perform active touch detection on the first capacitive sensing channel **103** (touch detection channel), the touch detection electrode **108** may comprise a “transmit pad” (or “drive pad”) on an IC die and the sense electrode **106** may comprise a “receive pad” (or “sense pad”) on the IC die, wherein signals (e.g., wave bursts) are transmitted by the touch detection circuitry **112** via the transmit pad (of the touch detection electrode **108**) and signals received at the receive pad (of the sense electrode **106**) are analyzed by the touch detection circuitry **112** for to perform touch detections **120** to detect a user touch, on a touch device (e.g., a touchscreen, touch button or switch, touch pad, touch slider, touch surface, or other touch device) including or connected to the IC die.
[0060] In some examples, the noise detection electrode **110** may comprise one or more passive (or “dummy”) conductive structures, e.g., a passive pad (“dummy pad”) on the IC die, a passive pin on an IC package in which the IC die is mounted, and/or a passive touch electrode element, without limitation. The noise detection circuitry **114** may detect noise-related capacitance measures **132** at the receive pad (of the sense electrode **106**), which noise-related capacitance measures **132** represent a passive noise-related capacitance between the noise detection electrode **110** and the receive pad. In other examples, the noise detection electrode **110** may comprise an active electrode, for example wherein the noise detection circuitry **114** transmits signals (e.g., wave bursts) on the noise detection electrode **110** and analyzes resulting signals received at the sense electrode **106** to perform noise detection measurement(s) **130** to detect the noise condition **134**.
[0061] FIG. **3** shows an example electronic device **300** representing another example implementation of the electronic device **100** shown in FIG. **1**. The example electronic device **300** may be generally similar to the example electronic device **200** shown in FIG. **2**, with like references numbers referring to like parts. However, in the example electronic device **300** shown in FIG. **3**, the second capacitive sensing channel **105** for noise detection is distinct from the first

capacitive sensing channel **103** for touch detection, as opposed to second capacitive sensing channel **105** and first capacitive sensing channel **103** sharing a sense electrode **106** (as in the example electronic device **200** discussed above.)

[0062] As shown in FIG. **3**, the first capacitive sensing channel **103** connected to the touch detection circuitry **112** includes the touch detection electrode **108** capacitively coupled to a first sense electrode **202** (e.g., corresponding with the sense electrode **106** of example electronic device **200** discussed above), and the second capacitive sensing channel **105** connected to the noise detection circuitry **114** includes the noise detection electrode **110** capacitively coupled to a second sense electrode **204**. The second sense electrode **204** may be separate from the first sense electrode **202**, and may comprise, for example, conductive elements of an IC die (e.g., an IC die pad and/or conductive trace), (b) a pin of an IC package in which the IC die is mounted, and/or I a respective touch electrode element.

[0063] FIG. **4** shows an example electronic device **400** representing an example implementation of the example electronic device **200** shown in FIG. **2** and discussed above, wherein the example electronic device **400** is embodied in an IC package **402** including an IC die **404**, and optionally a touch device **430**, for example a touchscreen (e.g., provided in a smartphone, laptop, tablet, or other device), a touch button or switch, a touch pad, a touch slider, a touch surface, or other touch device. The example electronic device **400** includes (a) the touch detection system **102** including the first capacitive sensing channel **103** and touch detection circuitry **112** to perform touch detections **120** (discussed above) based on the capacitance on the first capacitive sensing channel **103**, and (b) the noise detection system **104** including the second capacitive sensing channel **105** and noise detection circuitry **114** to detect noise conditions and control the operation of the electronic device **400** in response to detected noise conditions, e.g., as discussed above.

[0064] As shown in FIG. **4**, the first capacitive sensing channel **103** connected to the touch detection circuitry **112** includes the touch detection electrode **108** capacitively coupled to the sense electrode **106**, and the second capacitive sensing channel **105** connected to the noise detection circuitry **114** includes the noise detection electrode **110** capacitively coupled to the sense electrode **106**. Thus, the sense electrode **106** is shared by the first capacitive sensing channel **103** and the second capacitive sensing channel **105**, e.g., as discussed above regarding the example electronic device **200**.

[0065] The sense electrode **106**, the touch detection electrode **108**, and the noise detection electrode **110** may include respective elements of the IC package **402**, the IC die **404** mounted therein, and/or the optional touch device **430**.

[0066] In the example shown in FIG. **4**, the sense electrode **106** may include (a) a first die pad **412** (e.g., "receive pad") of the IC die **404**, (b) a first package pin **422** of the IC package **402**, and (optionally) (c) a first touch electrode element **432** provided in the optional touch device **430**. The first die pad **412** is connected (e.g., wire bonded) to the first package pin **422**, which is connected to the first touch electrode element **432**.

[0067] The touch detection electrode **108** may include (a) a second die pad **414** (e.g., "transmit pad") of the IC die **404**, (b) a second package pin **424** of the IC package **402**, and (optionally) (c) a second touch electrode element **434** provided in the optional touch device **430**. The second die pad **414** is connected (e.g., wire bonded) to the second package pin **424**, which is connected to the second touch electrode element **434**. The second touch electrode element **434** is capacitively coupled to the first touch electrode element **432**, e.g., for detecting a touch at a respective location on the touch device **430**.

[0068] The noise detection electrode **110** may include a third die pad **416** (e.g., "dummy pad") and/or optional conductive trace(s) **418** formed in the IC die **404**. The noise detection electrode **110** is capacitively coupled to the first touch electrode element **432** (in particular, to the first die pad **412**), e.g., for detecting noise in the vicinity of the electronic device **400**. The dimensions and/or location of the optional conductive trace(s) **418** relative to the third die pad **416** may be selected to provide a desired noise detection sensitivity.

[0069] In some examples, the IC die **404** may comprise a microcontroller. As noted above, the touch device **430** may comprise a touchscreen, touch button or switch, touch pad, touch slider, touch surface, or other touch device. The IC package **402** may be included in the touch device **430**, or otherwise connected to the touch device **430**.

[0070] FIG. **5** shows an example electronic device **500** representing another example implementation of the example electronic device **200** shown in FIG. **2**. The example electronic device **500** may be generally similar to the example electronic device **400** shown in FIG. **4**, with like references numbers referring to like parts. However, in the example electronic device **500** shown in FIG. **5**, the noise detection electrode **110** may include the third die pad **416** (e.g., "dummy pad") connected (e.g., wire bonded) to a third package pin **426** of the IC package **404**. The third package pin **426** (of the noise detection electrode **110**) is capacitively coupled to the first package pin **422** (of the sense electrode **106**), e.g., for detecting noise in the vicinity of the IC package **404**.

[0071] FIG. **6** shows an example electronic device **600** representing another example implementation of the example electronic device **200** shown in FIG. **2**. The example electronic device **600** may be generally similar to the example electronic devices **400** and **500** shown respectively in FIGS. **4** and **5**, with like references numbers referring to like parts. However, in the example electronic device **600** shown in FIG. **6**, the noise detection electrode **110** may include a third touch electrode element **436** capacitively coupled to the first touch electrode element **432** (of the sense electrode **106**), e.g., for detecting noise in the vicinity of the IC package **404**. As shown, the third touch electrode element **436** may be connected to the third package pin **426**, which is connected (e.g., wire bonded) to the third die pad **416** (e.g., "dummy pad").

[0072] The third touch electrode element **436** (of the noise detection electrode **110**) may be located remotely from the first touch electrode element **432** (of the sense electrode **106**), or otherwise located or arranged in the touch device **430** to prevent touch-based capacitive coupling between the third touch electrode element **436** and first touch electrode element **432**.

[0073] FIG. **7** shows an example electronic device **700** representing another example implementation of the example electronic device **200** shown in FIG. **2**. The example electronic device **700** may be generally similar to the example electronic devices **400**, **500**, and **600** shown respectively in FIGS. **4-6**, with like references numbers referring to like parts. However, the example electronic device **700** shows a third capacitive sensing channel **702** including an additional touch detection electrode **704** capacitively coupled to the sense electrode **106**, wherein the additional capacitive sensing channel **702** is connected to the touch detection circuitry **112** to perform touch detections **120** based on a capacitance on the third capacitive sensing channel **702**. Thus, electronic device **700** includes at least two touch detection channels, namely the first capacitive sensing channel **103** and the third capacitive sensing channel **702**, sharing the sense electrode **106**.

[0074] As shown, the additional touch detection electrode **704** may include (a) a fourth die pad **710** (e.g., an additional "transmit pad") of the IC die **404**, (b) a fourth package pin **712** of the IC package **402**, and (c) a fourth touch electrode element **712** provided in the optional touch device **430**. The fourth die pad **710** is connected (e.g., wire bonded) to the fourth package pin **712**, which is connected to the fourth touch electrode element **712**. The fourth touch electrode element **712** is capacitively coupled to the first touch electrode element **432**, e.g., for detecting a touch at a respective location on the touch device **430**.

[0075] It should be understood that any of the example electronic devices disclosed herein may include any number of touch detection channels (e.g., one, two, or more touch detection channels), wherein respective touch detection channels may (or may not) share a sense electrode with one or more other touch detection channels.

[0076] Further, the noise detection electrode **110** may include the third touch electrode element **436** capacitively coupled to the first touch electrode element **432** (of the sense electrode **106**), e.g., for

detecting noise (by noise detection circuitry **114**) in the vicinity of the IC package **404**.
[0077] FIG. **8** shows an example method **800** for managing capacitive touch detection. The example method **800** may be performed by any of the example devices disclosed herein.
[0078] At **802**, a touch detection circuitry provided in an IC device may perform at least one first touch detection based on a capacitance on a first capacitive sensing channel. For example, the touch detection circuitry may transmit signals via a transmit pad on an IC die, receive resulting signals on a receive pad on the IC die, and determine the presence of a user touch based on the received signals.
[0079] At **804**, a noise detection circuitry provided in the IC device may perform at least one noise detection measurement based on a capacitance on a second capacitive channel. The second capacitive channel may or may not share an electrode with the first capacitive channel, depending on the particular implementation, as discussed above. In some examples, performing a respective noise detection measurement comprises determining a respective noise-related capacitance measure (e.g., voltage, current, or other electrical parameter) associated with a capacitance on the second capacitive channel.
[0080] At **806**, the noise detection circuitry may determine a presence (or absence) of a noise condition based on the at least one noise detection measurement at **804**. For example, the noise detection circuitry may compare one or more noise-related capacitance measures (e.g., voltage, current, or other electrical parameter(s)) with at least one noise detection threshold to determine the presence (or absence) of the noise condition.
[0081] At **808**, in response to determining the presence of the noise condition, the noise detection circuitry may inhibit at least one second touch detection. (Alternatively, if the noise condition is determined to be absent at **806**, the noise detection circuitry may allow touch detections to continue, without interruption). For example, the noise detection circuitry may prevent a performance of at least one second touch detection by the touch detection circuitry. As another example, the noise detection circuitry may prevent a predefined number of second touch detections or prevent second touch detections for a predefined period of time. As another example, the noise detection circuitry may prevent second touch detections until a determination, by the noise detection circuitry, that the noise condition is absent. For example, a determination that the noise condition is absent may comprise a determination, by the noise detection circuitry, that the noise condition is not present for a predefined minimum number of noise detection measurements. As still another example, the noise detection circuitry may prevent or inhibit a use of at least one second touch detection performed by the touch detection circuitry.
[0082] FIG. **9** shows an example method **900** for managing capacitive touch detection. The example method **900** may be performed by any of the example devices disclosed herein.
[0083] At **902**, a touch detection circuitry may perform at least one first touch detection based on a capacitance on a first capacitive sensing channel, e.g., as discussed above.
[0084] At **904**, a noise detection circuitry may perform at least one noise detection measurement based on a capacitance on a second capacitive channel, e.g., as discussed above.
[0085] At **906**, the noise detection circuitry may determine whether a noise condition is present, based on the at least one noise detection measurement at **904**, e.g., as discussed above.
[0086] If the noise detection circuitry determines a noise condition is present at **906**, the noise detection circuitry at **908** may set a “noise detect” flag indicating the noise condition, and initiate or reset a debounce counter at **908**, by setting a value C of the debounce counter to N (i.e., setting C=N), wherein N represents a number of consecutive determinations (at **904-906**) that the noise detection is absent (no longer present) before returning to **902** to resume touch detections.
[0087] After setting the “noise detect” flag and setting the debounce counter C=N at **908**, the noise detection circuitry may inhibit touch detection(s) at **910** and decrement the debounce counter C at **912** (i.e., C=C−1), and the method may proceed to the next noise detection determination at **904**.
[0088] Alternatively, if the noise detection circuitry determines at **906** that the noise condition is

absent (not present), the noise detection circuitry may determine at **914** whether the system is currently in a (previously initiated) debouncing process, i.e., by determining whether the debounce counter C>0. If the system is currently in a debouncing process (C>0), the method may continue to inhibit touch detection(s) at **910** and decrement the debounce counter C at **912** (i.e., C=C−1), and the method may proceed to the next noise detection determination at **904**. Alternatively, if the system is not currently in a debouncing process (i.e., debounce counter C=0), touch detections are allowed to continue at **916**, and the method may return to **902** for a next touch detection. It should be understood that the debounce process (including the example debounce counter) shown in FIG. **9** and discussed above represent one example implementation, and the debounce process and/or debounce counter may be otherwise implemented in any suitable manner.
[0089] Although example embodiments have been described above, other variations and embodiments may be made from this disclosure without departing from the spirit and scope of these embodiments.

Claims

1. An electronic device, comprising: a touch detection system including: a first capacitive sensing channel; and a touch detection circuitry to perform a series of touch detections based on a capacitance on the first capacitive sensing channel; and a noise detection system including: a second capacitive sensing channel; and a noise detection circuitry to: perform at least one noise detection measurement, wherein a respective noise detection measurement comprises determining a respective noise-related capacitance measure associated with a capacitance on the second capacitive channel; and determine a presence of a noise condition based on the at least one noise detection measurement; and in response to determining the presence of the noise condition, generate a touch detection inhibit signal to inhibit at least one touch detection in the series of touch detections.
2. The electronic device of claim 1, including: a sense electrode; a touch detection electrode; and a noise detection electrode; wherein the first capacitive sensing channel includes a capacitive coupling between the touch detection electrode and the sense electrode; and wherein the second capacitive sensing channel includes a capacitive coupling between the noise detection electrode and the sense electrode.
3. The electronic device of claim 2, wherein: performing a respective touch detection in the series of touch detections comprises detecting a touch-related capacitance measure at the sense electrode at a first time; and performing the respective noise detection measurement comprises detecting the respective noise-related capacitance measure at the sense electrode at a second time distinct from the first time.
4. The electronic device of claim 2, comprising an IC package including: an IC die including the touch detection circuitry, the noise detection circuitry, and a plurality of IC die pads including a first IC die pad, a second IC die pad, and a third IC die pad; and a plurality of IC package pins, including: a first IC package pin connected to the first IC die pad; and a second IC package pin connected to the second IC die pad; wherein the sense electrode includes the first IC die pad and the first IC package pin connected to the first IC die pad; wherein the touch detection electrode includes the second IC die pad and the second IC package pin connected to the second IC die pad; and wherein the noise detection electrode comprises the third IC die pad.
5. The electronic device of claim 4, wherein the noise detection electrode comprises the third IC die pad and a conductive trace extending from the third IC die pad.
6. The electronic device of claim 4, wherein: the plurality of IC package pins includes a third IC package pin connected to the third IC die pad; and the noise detection electrode comprises the third IC die pad and the third IC package pin connected to the third IC die pad.
7. The electronic device of claim 4, wherein: the sense electrode includes a first touch electrode

element connected to the first IC die pad; the touch detection electrode includes a second touch electrode element connected to the second IC die pad; and the noise detection electrode includes a third touch electrode element connected to the third IC die pad.

8. The electronic device of claim 1, including: a touch detection electrode; a first sense electrode; a noise detection electrode; and a second sense electrode; wherein the first capacitive sensing channel includes a capacitive coupling between the touch detection electrode and the first sense electrode; and wherein the second capacitive sensing channel includes a capacitive coupling between the noise detection electrode and the second sense electrode.

9. The electronic device of claim 1, wherein determining the presence of the noise condition based on the at least one noise-related capacitance measurement comprises comparing the at least one noise-related capacitance measure to a noise detection threshold.

10. The electronic device of claim 1, wherein generating the touch detection inhibit signal to inhibit the at least one touch detection comprises generating the touch detection inhibit signal to prevent a performance of the at least one touch detection.

11. The electronic device of claim 1, wherein generating the touch detection inhibit signal to inhibit the at least one touch detection comprises generating the touch detection inhibit signal to inhibit a usage of the at least one touch detection.

12. The electronic device of claim 1, wherein generating the touch detection inhibit signal to inhibit the at least one touch detection comprises generating the touch detection inhibit signal to inhibit a predefined multiple number of touch detections.

13. The electronic device of claim 1, wherein the noise detection circuitry to: perform at least one further noise detection measurement, wherein a respective further noise detection measurement comprises determining a respective further noise-related capacitance measure associated with the capacitance on the second capacitive channel; determine an absence of the noise condition based on the at least one further noise detection measurement; and in response to the determination of the absence of the noise condition, not generate the touch detection inhibit signal.

14. The electronic device of claim 1, wherein the touch detection circuitry to perform a respective touch detection in the series of touch detections by: determining a capacitance measure associated with a capacitance on the first capacitive sensing channel; and comparing the touch-related capacitance measurement to a touch detection threshold.

15. The electronic device of claim 1, wherein: the device comprises a touch device including a first touch electrode element and a second touch electrode element; and the first capacitive sensing channel includes a capacitive coupling between the first touch electrode element and the second touch electrode element.

16. The electronic device of claim 1, comprising: an IC die including the touch detection circuitry and the noise detection circuitry; a touch device including: a first touch electrode element connected to a first IC pad of the IC die; and a second touch electrode element connected to a second IC pad of the IC die and capacitively coupled to the first touch electrode element; wherein the first touch electrode element and the second touch electrode element are arranged along the first capacitive sensing channel; and wherein the touch detection circuitry to perform a series of touch detections based on a capacitive coupling between the first touch electrode element and the second touch electrode element.

17. A method, comprising: performing, using a touch detection circuitry provided in an integrated circuit, at least one first touch detection based on a capacitance on a first capacitive sensing channel; performing, using a noise detection circuitry provided in the IC device, at least one noise detection measurement based on a capacitance on a second capacitive channel; determining, by the noise detection circuitry, a presence of a noise condition based on the at least one noise detection measurement; and in response to determining the presence of the noise condition, inhibiting at least one second touch detection.

18. The method of claim 17, wherein: performing the at least one noise detection measurement comprises determining at least one noise-related capacitance measure associated with the capacitance on the second capacitive channel; and determining the presence of the noise condition comprises comparing the at least one noise-related capacitance measure to a noise detection threshold.

19. The method of claim 17, wherein inhibiting at least one second touch detection comprises preventing a performance of at least one second touch detection by the touch detection circuitry.

20. The method of claim 19, wherein inhibiting at least one second touch detection comprises preventing a predefined number of second touch detections or preventing second touch detections for a predefined period of time.

21. The method of claim 19, wherein inhibiting at least one second touch detection comprises preventing second touch detections until a determination, by the noise detection circuitry, that the noise condition is absent.

22. The method of claim 21, wherein the determination, by the noise detection circuitry, that the noise condition is absent comprises a determination, by the noise detection circuitry, that the noise condition is not present for a predefined minimum number of noise detection measurements.

23. The method of claim 17, wherein inhibiting at least one second touch detection comprises preventing or inhibiting a use of at least one second touch detection performed by the touch detection circuitry.

24. The method of claim 17, wherein: performing at least one first touch detection comprises detecting at least one first touch at a touch device; and inhibiting at least one second touch detection comprises (a) inhibiting a detection of at least one second touch at the touch device or (b) inhibiting a use of at least one detected second touch at the touch device.